



Technische Hochschule
Ingolstadt

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Autonomes Fliegen

Kapitel 6: Sensors and Inertial Navigation

WS 2025/26

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10.12.2025

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Content

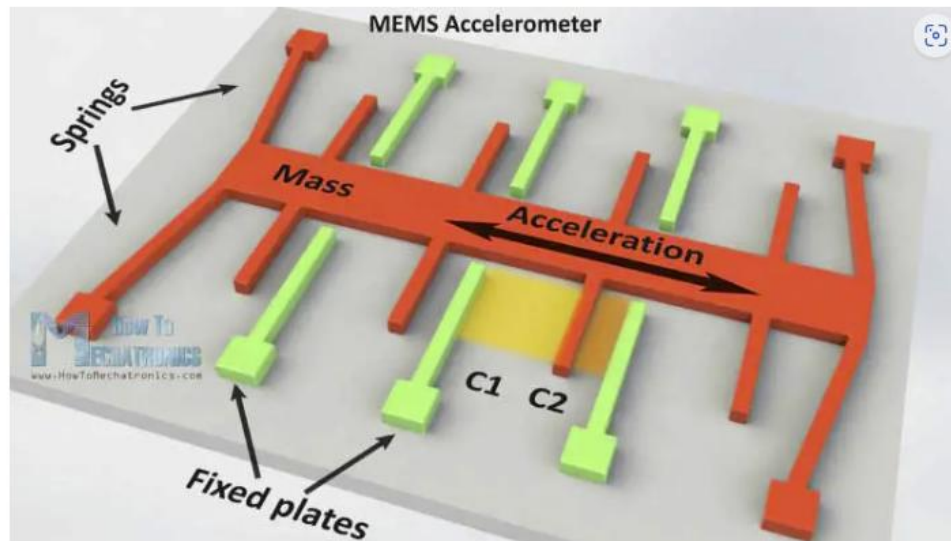
- **Sensors**
 - Flight- & Navigation Sensors
 - Accelerometers
 - Gyroscopes
 - GPS, Baro, Magnetometer / Northfinder, Air-Data
 - Environmental Sensors
 - Camera,
 - LiDAR,
 - Radar,
- **Strapdown inertial navigation**

Sensors

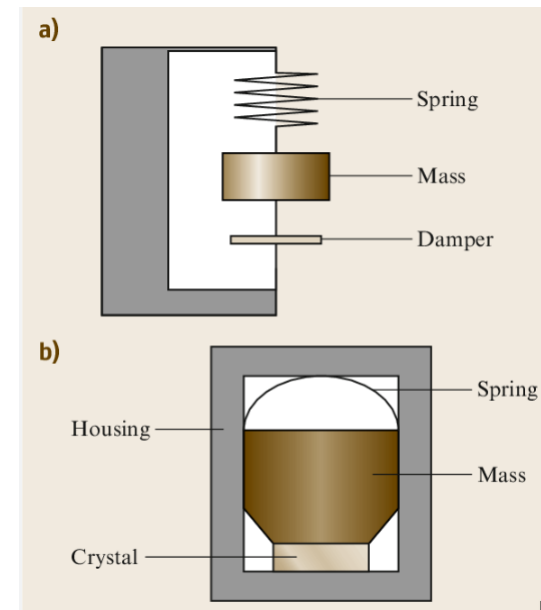
Sensors – Flight- & Navigation Sensors

Accelerometers

- Measures all external forces acting upon them (including gravity)
- Acts like a spring-damper system
- To obtain inertial acceleration (due to motion alone), gravity must be subtracted



MEMS Accelerometer



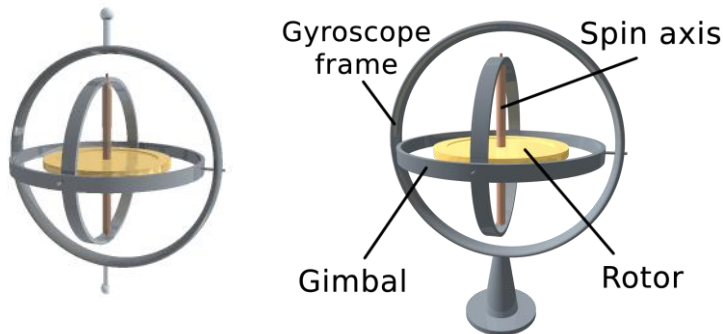
Accelerometers (a) Mechanical accelerometer.
(b) Piezoelectric accelerometer

Sensors – Flight- & Navigation Sensors

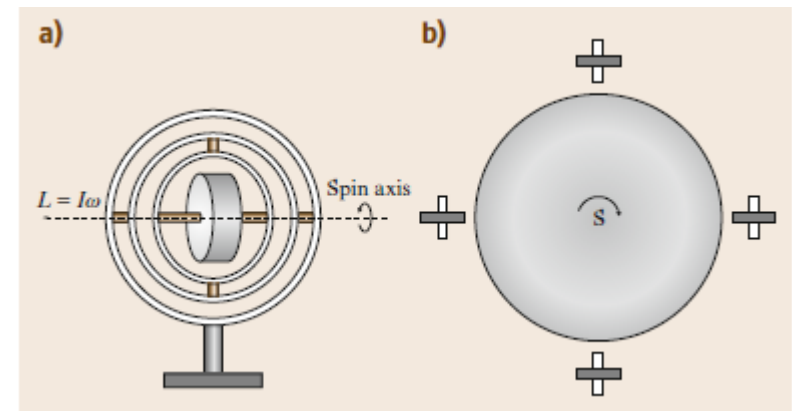


Gyroscopes

- **Measures orientation** (standard gyro) or (rate gyro, needs integration for angle)
- Spinning wheel mounted in a gimbal device (can move freely in 3 dimension)
- Wheel keeps orientation due to angular momentum



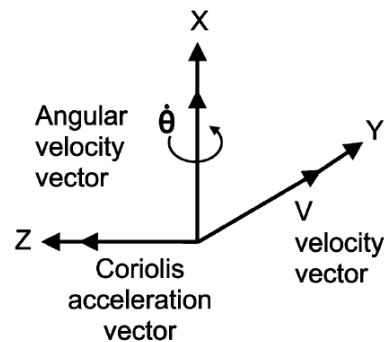
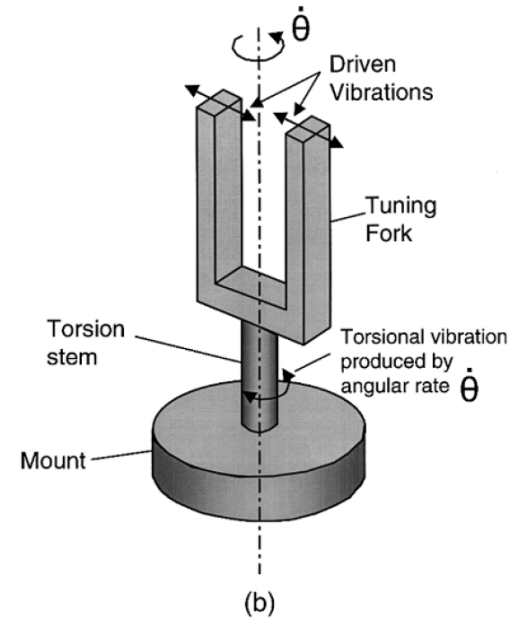
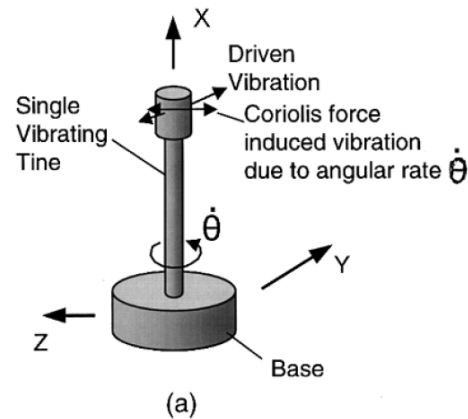
https://upload.wikimedia.org/wikipedia/commons/d/d5/Gyroscope_operation.gif



$$L = I \times \omega$$

Sensors – Flight- & Navigation Sensors

Gyroscopes



$$\text{Coriolis acceleration} = 2 \mathbf{V} \times \dot{\theta}$$

Velocity of time, V , vibrating at ω rads./s and amplitude, A , is:

$$V = A \omega \cos \omega t$$

$$\text{Coriolis accln.} = 2 A \omega \cos \omega t \dot{\theta}$$

$$\text{Coriolis force on tine of mass, } m = 2 m A \omega \dot{\theta} \cos \omega t$$

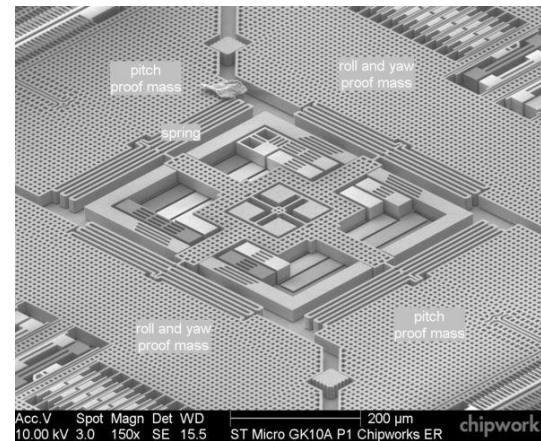
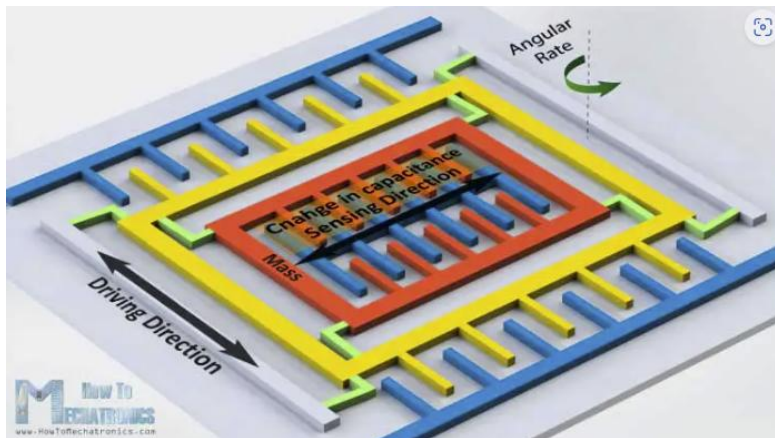
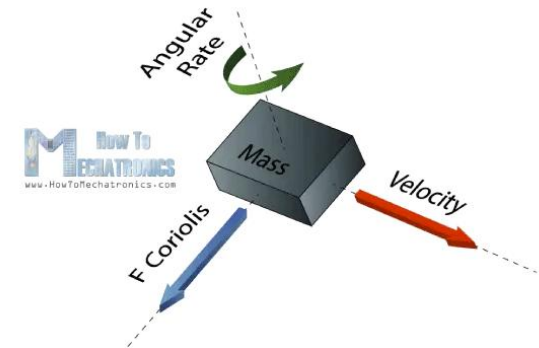
Hence, amplitude of induced vibration = $K \dot{\theta}$. (K = constant)

Sensors – Flight- & Navigation Sensors

MEMS Gyroscopes



- **Vibrating structure gyroscope (MEMS)**
 - Based on Coriolis effect
 - „Vibration keeps its direction under rotation“
 - Implementations: **Tuning fork**, vibration wheels, ...



https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.memsjournal.com%2F2011%2F01%2Fmotion-sensing-in-the-iphone-4-mems-gyroscope.html&psig=AOvVaw3_4VN6MmSSs40wRgNQd1vE&ust=1699869926457000&source=images&cd=vfe&ved=0CBEQjRxqFwoTCljfvIqbv oIDFQAAAAAdAAAAABAs

Sensors – Flight- & Navigation Sensors

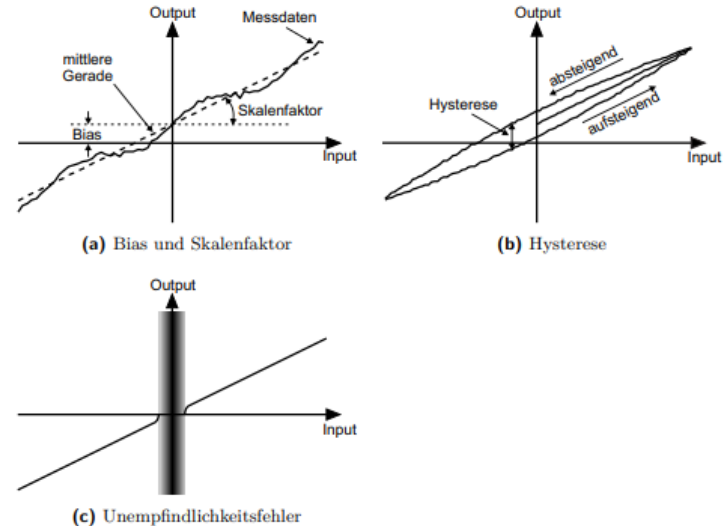


Fehlerklassen/Größenordnungen/Wichtige Sensorfehler/Requirement

Einordnung/Kategorien

Kategorie	Qualitätskriterium	Größenordnung
strategic grade	Gyro Bias Stabilität	$< 0.0001 \text{ } ^\circ/\text{h}$
	Gyro Skalenfaktor Stabilität	$< 50 \text{ ppm}$
	Accelerometer Bias Stabilität	$< 1 \text{ } \mu\text{g}$
	Accelerometer Skalenfaktor Stabilität	$< 2 \text{ ppm}$
navigation grade	Gyro Bias Stabilität	$0.0001 - 0.1 \text{ } ^\circ/\text{h}$
	Gyro Skalenfaktor Stabilität	$1 - 100 \text{ ppm}$
	Accelerometer Bias Stabilität	$1 - 1000 \text{ } \mu\text{g}$
	Accelerometer Skalenfaktor Stabilität	$1 - 100 \text{ ppm}$
tactical grade	Gyro Bias Stabilität	$0.1 - 10000 \text{ } ^\circ/\text{h}$
	Gyro Skalenfaktor Stabilität	$> 100 \text{ ppm}$
	Accelerometer Bias Stabilität	$50 - 10000 \text{ } \mu\text{g}$
	Accelerometer Skalenfaktor Stabilität	$> 100 \text{ ppm}$

Sensorfehler



Performance Requirements

		Flight control system	Strapdown INS
Gyro	Scale factor	0.5 %	0.001% (10 ppm)
	Zero offset/rate uncertainty	$1^\circ/\text{min}$	$0.01^\circ/\text{hour}$
Acctr	Scale factor	0.5%	0.01% (100 ppm)
	Zero offset/bias stability	$5 \times 10^{-3} \text{ g}$	$5 \times 10^{-5} \text{ g}$ (50 μg)

Bias ist laut Definition ein **zeitlich konstanter** Fehler

- Ist der Wert, der vom Sensor bei **fehlendem Eingangssignal** ausgegeben wird
- Physikalische Effekte die zu einem Bias führen
 - Herstellungsprozess/Fertigungstoleranzen
 - Temperaturabhängigkeit
 - Langzeitstabilität
 - Einsatzumgebung (z.B. Vibration, Elektromagnetische Interferenzen)
- **Größenordnung-B-Messer: Typische Bias-Werte können im Bereich von wenigen Millig (m/s^2) bis zu einigen Zehntel g (Erdbeschleunigung) liegen, abhängig von der Qualität des Sensors.**
- **Bias-Gyros:** Es ist der Wert der auftritt obwohl **keine Drehbewegung** auftritt.
 - In der Fachliteratur wird beim Gyro der Begriff Bias oft als **Drift** bezeichnet.
 - Grund: Bei freien mechanischen Kreiseln ändert sich die Laufachse aufgrund des fortlaufend auftretenden Bias mit der Zeit kontinuierlich und führt somit zu einer Drift.
 - Größenordnung (typische UAV Sensoren): **Bias-Werte können im Bereich von wenigen Grad pro Sekunde ($^\circ/\text{s}$) bis zu mehreren Zehntel Grad pro Sekunde liegen, abhängig von der Qualität des Sensors.**
- Um mit Bias in MEMS-Gyroskopen umzugehen, sind Kalibrierungsverfahren notwendig.
 - Diese Verfahren **erfassen** und **korrigieren** die bekannten Abweichungen
 - Eine genaue Kalibrierung berücksichtigt verschiedene Umgebungsbedingungen und Betriebszustände, um zuverlässige und präzise Messungen sicherzustellen.

- **Skalenfaktor:** beschreibt Verhältnis zwischen Eingangs- und Ausgangssignal
 - Viele Sensoren weisen Ausgangssignal auf, das **direkt proportional** zum Eingangssignal ist
 - Der Skalenfaktor entspricht somit Steigung jener Geraden, die man erhält, wenn man das Eingangssignal über einen entsprechenden Bereich variiert
 - Die Steigung der mittleren Geraden erhält man mittels Ausgleichsgerade
 - Im Falle von Nichtlinearitäten des Skalenfaktors kann ein Polynom höherer Ordnung angepasst werden
- **Skalenfaktorfehler-MEMS B-Messer:** entspricht bei einem **linearen Skalenfaktor** der **maximal** möglichen **Abweichung** von der im Ausgleich bestimmten Geraden und wird oft als Verhältniszahl der **maximalen Abweichung** zum Messintervall angegeben.
- Größenordnung (typische UAV Sensoren): Skalierungsfehler können im Bereich von **wenigen Prozent** der gemessenen Beschleunigung liegen.

■ Rauschen (Noise):

- Beschreibung: Rauschen ist zufällige Schwankung in den gemessenen Werten. Es kann durch elektronische Störungen, thermisches Rauschen oder andere Umgebungsbedingungen verursacht werden.
- Größenordnung: Das Rauschen kann im Bereich von **Mikrog (μg) bis Millig (m/s^2)** liegen, abhängig von der Qualität des Sensors und den Umgebungsbedingungen.

■ Linearitätsfehler

- Beschreibung: Der Linearitätsfehler tritt auf, wenn die tatsächliche Beziehung zwischen der Eingangsbeschleunigung und der Ausgangsspannung oder digitalen Werte nicht genau linear ist.
- Größenordnung: Die typischen Größenordnungen von Linearitätsfehlern können im Bereich von Bruchteilen eines Prozent der gemessenen Beschleunigung liegen.

■ Temperaturdrift

- Beschreibung: Die Empfindlichkeit des Sensors gegenüber Temperaturschwankungen kann zu Drift in den Messungen führen.
- Größenordnung: Temperaturdrift kann im Bereich von Mikrograd pro Sekunde pro Grad Celsius ($\mu^\circ/\text{s}/^\circ\text{C}$) bis zu Milligrad pro Sekunde pro Grad Celsius liegen.

MEMS- Gyros

■ Rauschen (Noise):

- Beschreibung: Rauschen ist zufällige Schwankung in den gemessenen Werten und kann durch elektronische Störungen, thermisches Rauschen oder andere Umgebungsbedingungen verursacht werden.
- Größenordnung: Das Rauschen kann im Bereich von Mikrograd pro Sekunde (μ°/s) bis zu Milligrad pro Sekunde liegen, abhängig von der Qualität des Sensors und den Umgebungsbedingungen.

■ Linearitätsfehler:

- Beschreibung: Der Linearitätsfehler tritt auf, wenn die tatsächliche Beziehung zwischen der gemessenen Winkelgeschwindigkeit und der Ausgangsspannung oder digitalen Werte nicht genau linear ist.
- Größenordnung (typische UAV Sensoren): Linearitätsfehler können im Bereich von Bruchteilen eines Prozent der gemessenen Winkelgeschwindigkeit liegen.

■ Temperaturdrift

- Beschreibung: Die Empfindlichkeit des Sensors gegenüber Temperaturschwankungen kann zu Drift in den Messungen führen.
- Größenordnung (typische UAV Sensoren): Temperaturdrift kann im Bereich von Mikrograd pro Sekunde pro Grad Celsius ($\mu^\circ/\text{s}/^\circ\text{C}$) bis zu Milligrad pro Sekunde pro Grad Celsius liegen.

■ Kreuzkopplungsfehler

- Er tritt vor allem bei Sensoransammlungen (z. B. einer IMU), aber auch bei einzelnen Sensoren auf.
- Er gibt an, welche Fehler durch Instabilitäten der sensitiven Achsen entstehen.
- Bei einer IMU beschreibt er jenen Fehler, der auftritt, wenn die Sensorachsen nicht aufeinander orthogonal stehen.

■ Random Drift

- Beim Random Drift befindet sich der Sensor in Ruhe (kein Signal am Eingang),
- Messwert am Ausgang variiert jedoch aufgrund von Störungen innerhalb des Sensors um kleine Beträge

■ Random Walk

- Fehler, der z. B. bei optischen Gyros auftritt.
- Das weiße Messrauschen führt auf lange Zeit gesehen zu einem **zufälligen** Anwachsen von Fehlern in der Attitude.

2 Mechanical and electrical specifications

2.1 Mechanical characteristics

@ Vdd = 3.0 V, T = 25 °C unless otherwise noted^(a).

Table 3. Mechanical characteristics

Symbol	Parameter	Test condition	Min.	Typ. ⁽¹⁾	Max.	Unit
FS	Measurement range	User selectable		±245 ±500 ±2000		dps
So	Sensitivity			8.75 17.50 70.00		mdps/digit
SoDr	Sensitivity change vs. temperature ⁽²⁾	From -40 °C to +85 °C Delta from T = 25 °C		±2		%
DVoff	Digital Zero-rate level	FS = 2000 dps		±25		dps
OffDr	Zero-rate level change vs temperature ⁽³⁾	FS = 2000 dps		±0.04		dps/°C
NL	Non linearity ⁽³⁾	Best fit straight line		0.2		% FS
Rn	Rate noise density ⁽³⁾	BW = 50 Hz		0.011		dps/(√Hz)
ODR	Digital output data rate ⁽³⁾			11.9/23.7/ 47.3/94.7/ 189.4/ 378.8/ 757.6		Hz
Top	Operating temperature range		-40		+85	°C

1. Typical specifications are not guaranteed.

2. Guaranteed by design.

3. The period (1/ODR), length of time between two consecutive sampling, must be derived by the reciprocal of the maximum and minimum ODR limits: for example for ODR = 189.4 Hz, sampling period range will be within [4591 μs, 6211 μs] (where ODR minimum and maximum have been approximated at 162 Hz, 219 Hz respectively).



L3GD20H

MEMS motion sensor:
three-axis digital output gyroscope

Datasheet - production data



LGA-16 (3x3x1 mm)

Applications

- Gaming and virtual reality input devices
- Motion control with MM (man-machine interface)
- GPS navigation systems
- Appliances and robotics

Description

The L3GD20H is a low-power three-axis angular rate sensor.

It includes a sensing element and an IC interface able to provide the measured angular rate to the external world through digital interface (I2C/SPI).

The sensing element is manufactured using a dedicated micromachining process developed by ST to produce inertial sensors and actuators on silicon wafers.

The IC interface is manufactured using a CMOS process that allows a high level of integration to design a dedicated circuit which is trimmed to better match the sensing element characteristics.

The L3GD20H has a full scale of ±245/±500/±2000 dps and is capable of measuring rates with a user selectable bandwidth.

The L3GD20H is available in a plastic land grid array (LGA) package and can operate within a temperature range from -40 °C to +85 °C.

Features

- Wide supply voltage, 2.5 V to 3.6 V
- Wide extended operating temperature range (from -40 °C to +85 °C)
- Low voltage compatible I/Os, 1.8 V
- Low power consumption
- Embedded power-down
- Sleep mode
- Fast turn-on and wake-up
- Three selectable full scales up to 2000 dps
- 16 bit rate value data output
- 8 bit temperature data output
- I2C/SPI digital output interface
- 2 dedicated lines (1 interrupt, 1 data ready)
- User enable integrated high-pass filters
- Embedded temperature sensor
- Embedded 32 levels of 16 bit data output FIFO
- High shock survivability
- ECOPACK[®] RoHS and "Green" compliant

Table 1. Device summary

Order code	Temperature range (°C)	Package	Packing
L3GD20H	-40 to +85	LGA-16 (3x3x1)	Tray
L3GD20HTR	-40 to +85	LGA-16 (3x3x1)	Tape and reel

L3GD20H

Mechanical and electrical specifications

2.2 Electrical characteristics

@ Vdd = 3.0 V, T = 25 °C unless otherwise noted^(b).

Table 4. Electrical characteristics

Symbol	Parameter	Test condition	Min.	Typ. ⁽¹⁾	Max.	Unit
Vdd	Supply voltage		2.2	3.0	3.6	V
Vdd_IO	I/O pins supply voltage ⁽²⁾		1.71		Vdd+0.1	V
Idd	Supply current			5.0		mA
IddSL	Supply current in sleep mode ⁽³⁾	Selectable by digital interface		2.5		mA
IddPdn	Supply current in power-down mode	Selectable by digital interface		1		μA
VIH	Digital high level input voltage		0.8*Vdd_IO			V
VIL	Digital low level input voltage				0.2*Vdd_IO	V
Ton	Turn-on time ⁽⁴⁾	LPF2 disabled ODR = 190 Hz		50		ms
Top	Operating temperature range		-40		+85	°C

1. Typical specifications are not guaranteed.

2. It is possible to remove Vdd maintaining Vdd_IO without blocking the communication busses, in this condition the measurement chain is powered off.

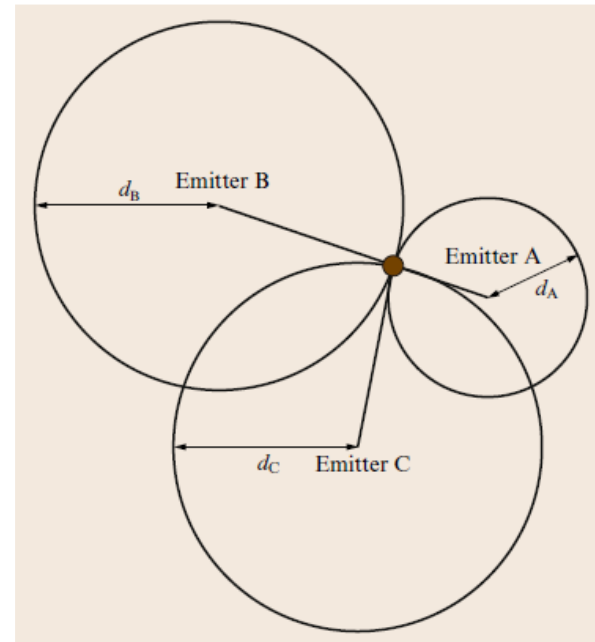
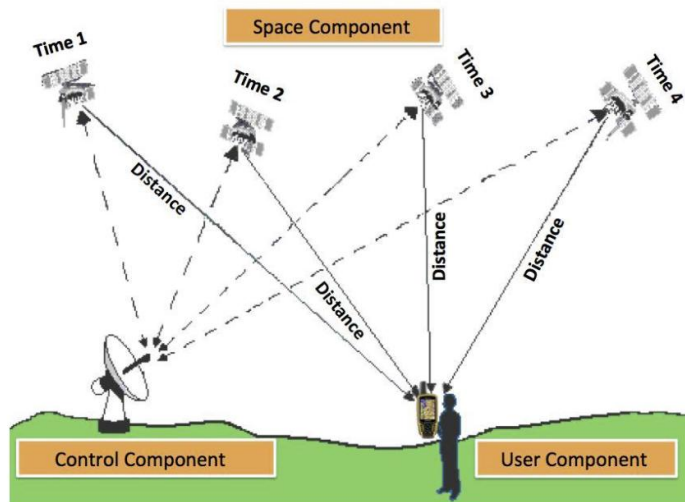
3. Sleep mode introduces a faster turn-on time related to power down mode.

4. Time to obtain stable sensitivity (within ±5% of final value) after exiting power-down mode. It is guaranteed by design.

Sensors – Flight- & Navigation Sensors

GPS (1/3)

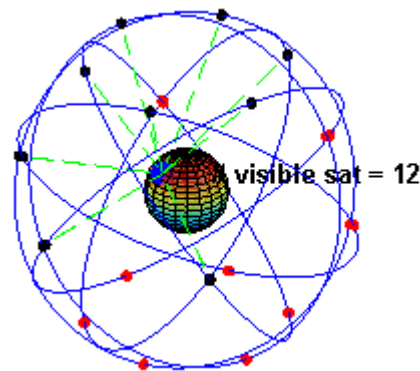
- Every satellite transmits **position** and **time**
- Receiver measures **time difference** of satellites signals
- Calculates position by **intersecting distances** (pseudoranges)



Sensors – Flight- & Navigation Sensors

GPS (2/3)

- 24+ satellites, 12 hour orbit, 20.190 km height
- 6 orbital planes, 4+ satellites per orbit, 60 deg distance
- Satellite transmits orbital location (almanach) + time
- 50 bits/s, msg has 1500 bits => **12.5 minutes**



<https://upload.wikimedia.org/wikipedia/commons/9/9c/ConstellationGPS.gif>

Sensors – Flight- & Navigation Sensors

GPS (3/3)

- Position from **pseudorange**
 - Requires measurements of 4 different satellites
 - Low accuracy (3-15m) but absolute
- Position from **pseudorange + phase shift** (RTK/dGPS)
 - Very precise ($< 1\text{mm}$)
 - Position: relative to a reference station

Sensors – Flight- & Navigation Sensors

Magnetic field sensor (Digital Compass)

- Measures **absolute** orientation in one axis
- Much less accuracy than gyroscope for **relative orientation**
- Easily affected by metal (door frames etc.)

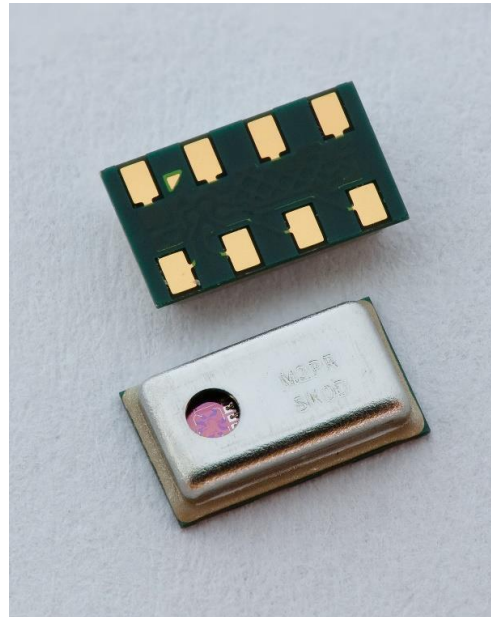


[Image source: Wikipedia, CC BY-SA 4.0]

Sensors – *Flight- & Navigation Sensors*

Air pressure sensor

- Measurement of air pressure can be used to measure altitude
- MEMS implementation
- Measurement depends highly on weather changes (temperature)
- Environment changes (open/closing doors or windows changes the measurement)

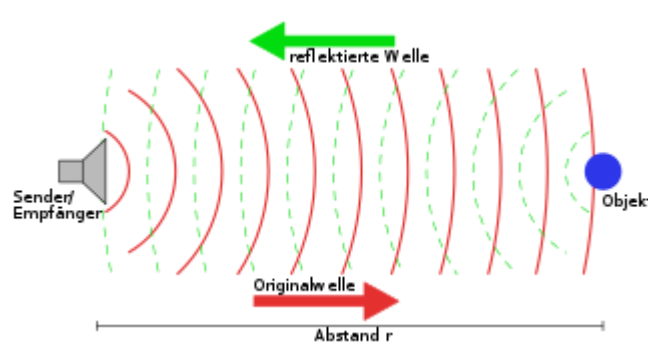


[Image source: Wikipedia, CC BY-SA 3.0]

Sensors - Environmental Sensors

Ultrasound Range Sensors (1/2)

- Emit signal to determine distance along a ray
- Make use of propagation speed of ultrasound
- Traveled distance is given by speed of sound ($v = 340 \text{ m/s}$)



$$d = \frac{v\Delta t}{2}$$

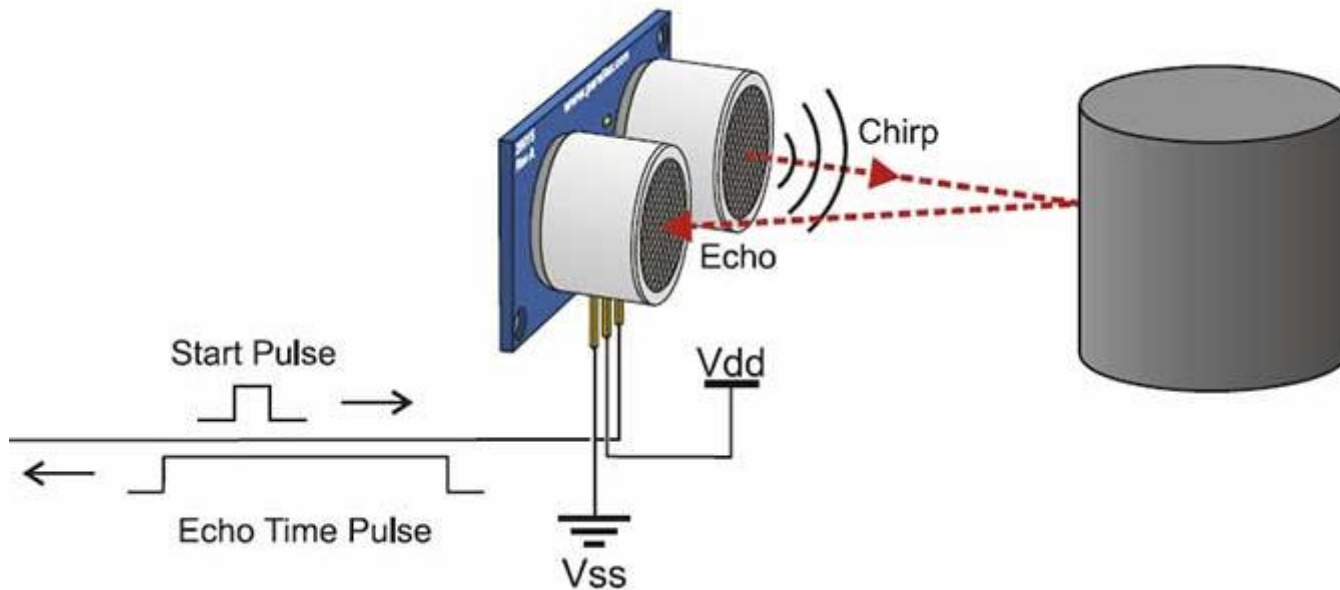
https://de.wikipedia.org/wiki/Datei:Sonar_Principle_DE.svg

Prinzip des aktiven Sonars: Die Zeit vom Aussenden des Impulses bis zum Empfangen des reflektierten Signals ist proportional zur Objektentfernung

Sensors - Environmental Sensors

Ultrasound Range Sensors (2/2)

- Range between 12cm and 5m
- Opening angle around 20 to 40 degrees
- Problems: multi-path propagation, absorption
- Lightweight and cheap



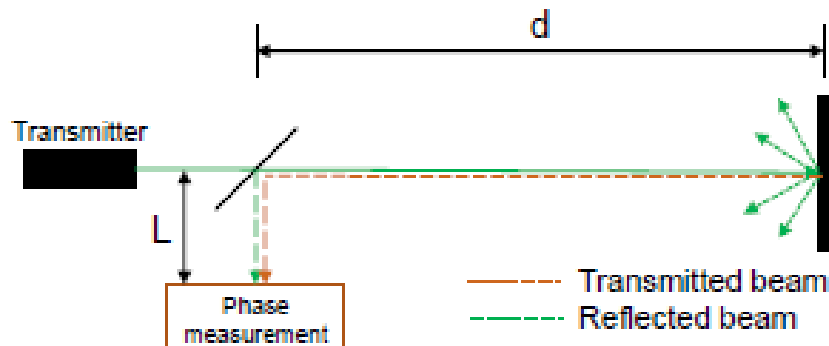
Sensors - Environmental Sensors

Laser distance sensor

Technical specifications

Product code	TR-EVO-15M
Performance	
Detection principle	Infrared Time-of-Flight
Range *	0.5m up to 15m
Update rate	Up to 240 readings per second
Output resolution	0.5cm below 14m, 2cm from 14m
Accuracy	±4cm in the first 14m, 1.5% above 14m
Field of View	Approx. 2°
Electronics	
Supply voltage	5V DC +/-5%
Supply current (min-max)	90mA - 330mA
Communication	
Interfaces	USB 2.0 Micro-B UART, +3.3V level, 115200,8,N,1 IC2, +3.3V level, 400kHz
Connectors	Single 9 pin Hirose DF13 Micro USB
Mechanical data	
Dimensions	Approx. 29x29x22mm (sensor + backboard)
Weight	9 g (sensor) + 3 g (backboard)
Eye safety	Yes (CE certified)

*Specifications are derived from tests in controlled conditions. Bright sunlight, target surface reflectivity and other variables will affect sensor performance. The combination of very bright sunshine and low reflectivity targets (such as grass) can reduce maximum range to 10m or less. For additional information, please find the following [test results report](#).



Quelle: terabee.com
TeraRanger Evo 15m - The medium-range ToF distance sensor, 15 m, 240 Hz, 12 grams (~ 60€)



Sensors - Environmental Sensors

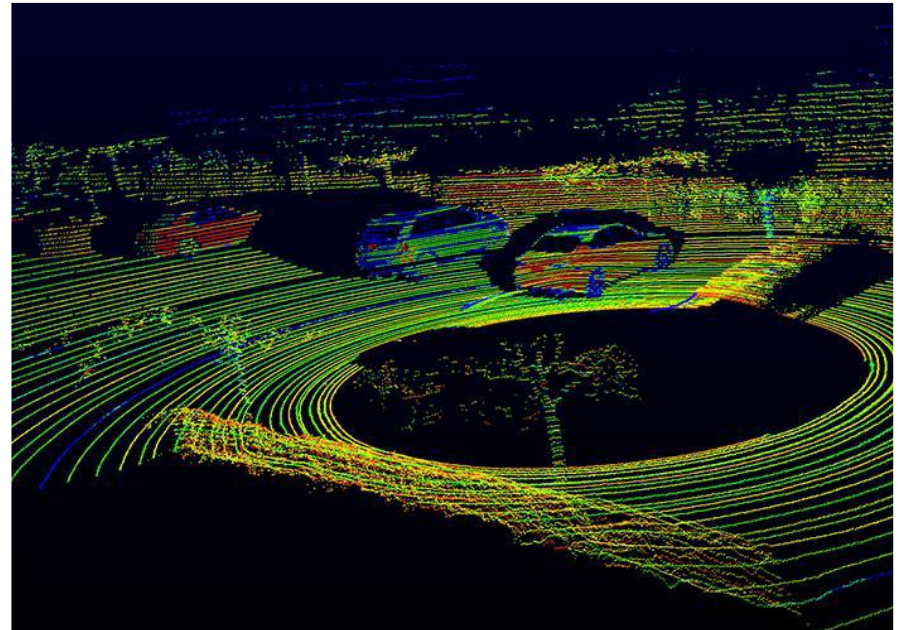
Laser range finder (Lidar)

- Rotating laser distance sensor (one axis or multiple axis)
- Line scanner or 3D scanner
- 3D scanner returns complex 3D point cloud



[Image source: velodynelidar.com]

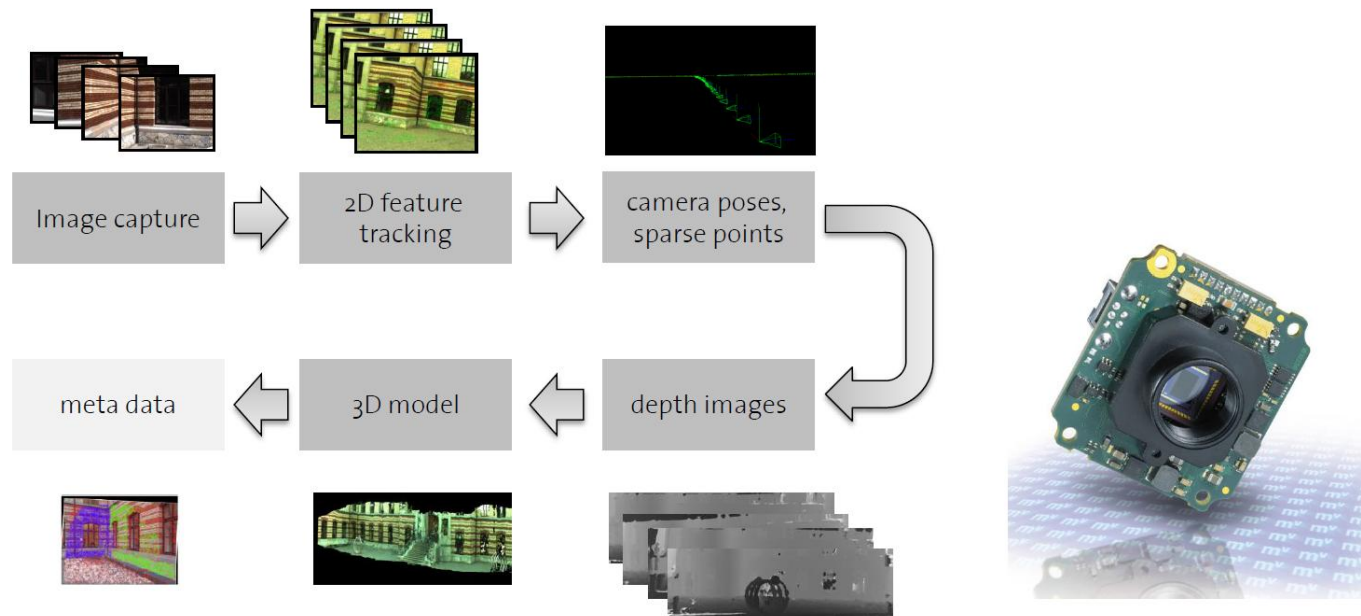
https://youtu.be/x7De3tCb3_A



Sensors - Environmental Sensors

Digital camera

- Versatile sensor
- Possibility to estimate full 6 DOF pose (orientation and position) and to obtain a 3D point cloud
- However, complex image processing algorithms necessary and necessity to take multiple images



Sensors - Environmental Sensors

Optical flow sensor

- Sensor used in computer mouse
- Low resolution infrared camera
- Measures x,yshift of infrared image
- Method needs very high update rates (image shift needs to be small)

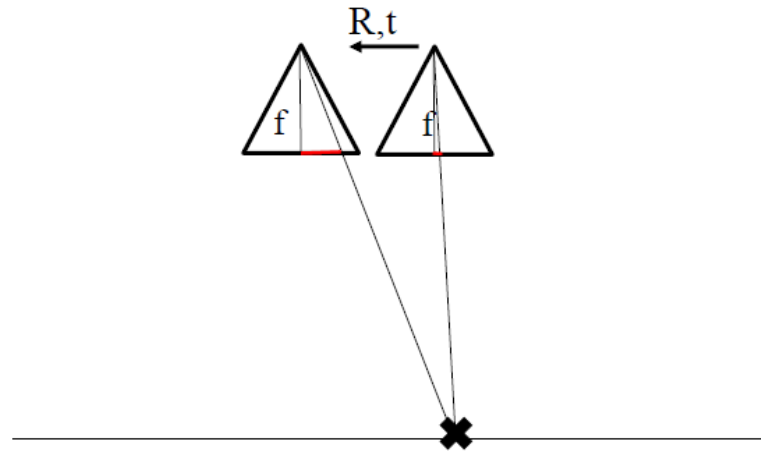
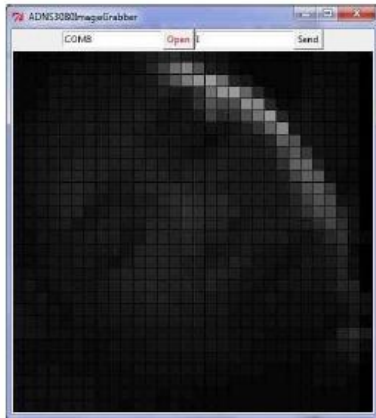


Image source ardupilot [\[Image source:\]](#)

Sensors - Environmental Sensors

Examples (1/2) Optical flow camera (PX4Flow)

- Smart camera measuring x,y,z movement in meters
- Combines **camera**, **gyroscope**, **ultrasound** and processor
- Processor computes optical flow from image with 200Hz
- Gyroscope computes optical flow component induced by tilting (to be removed from x,y measurement)
- Ultrasound sensor used to convert pixel measurement into meter

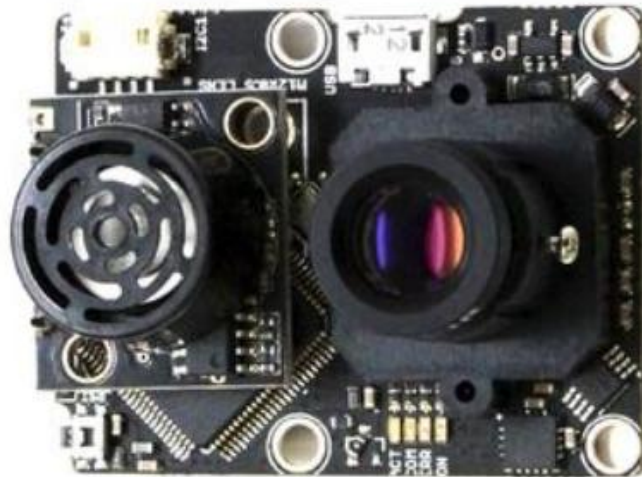


Image source [Holybro PX4FLOW KIT V1.31 - optischer Flow Sensor Kamera | Flight Controller & Zubehör | Fernsteuerung & Elektronik | Zubehör | Premium-Modellbau](#)

Strapdown Inertial Navigation

Strapdown inertial navigation

Inertial Measurement Units (IMUs)

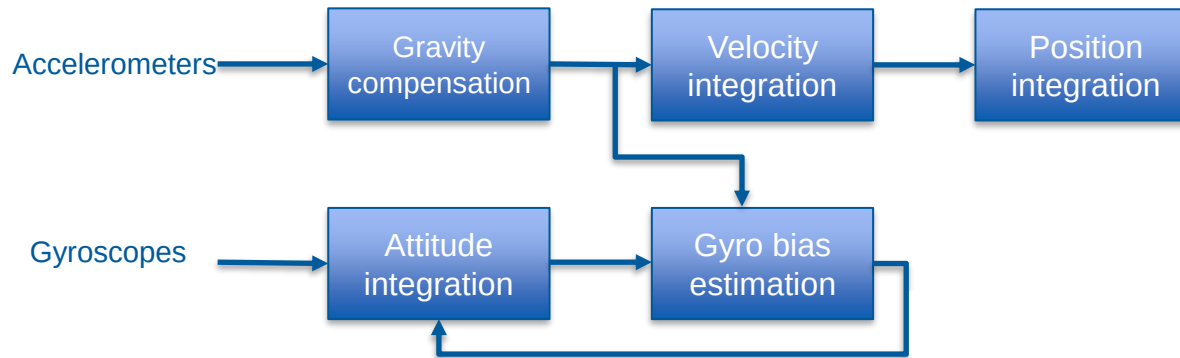
- **3-axis MEMS gyroscope**
 - Provides angular velocity
 - Integrate for angular position
 - Problem: Drifts slowly over time (e.g. 1 deg/hour), called the bias
- **3-axis MEMS accelerometer**
 - Provides accelerations (including gravity)

References

1. Britting, K. R., *Inertial Navigation Systems Analysis*, John Wiley & Sons, Inc., New York (1971). Reprinted by Artech House in 2010.
2. Titterton, D. H., and Weston, J. L., *Strapdown Inertial Navigation Technology*, 2nd ed., The Institution of Electrical Engineers, Stevenage, United Kingdom (2004).

Strapdown inertial navigation

IMU Strapdown Algorithm (1/4) - Overview



IEEE SENSORS JOURNAL, VOL. 10, NO. 4, APRIL 2010

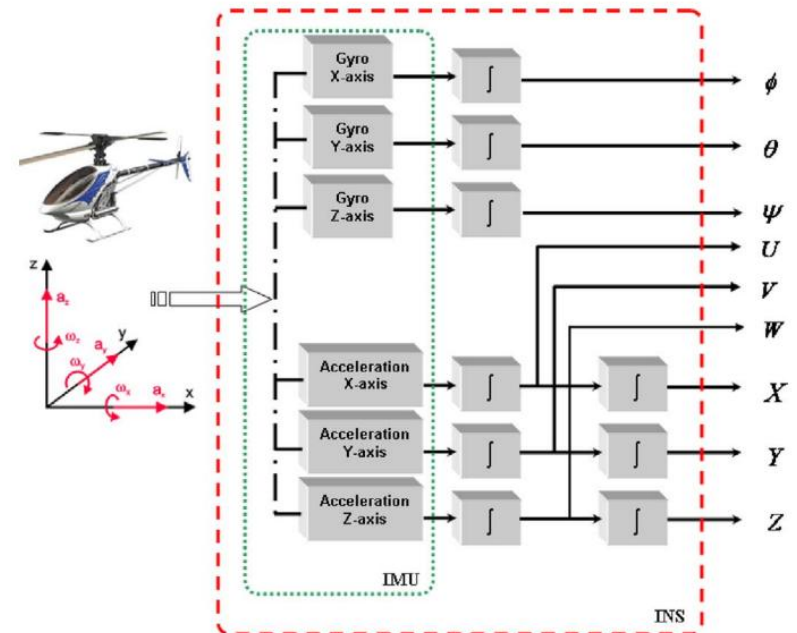


Fig. 1. INS architecture.

Strapdown inertial navigation

IMU Strapdown Algorithm (2/4) – Signals from an inertial measurement unit

Referring to Figure 1, suppose the motion of a rigid body is characterized by a rotation tensor $\mathbf{Q}$ and the position vector $\mathbf{x}_A$ of a point A on the body where an IMU is attached. The point A is sometimes called a landmark. The rotation tensor $\mathbf{Q}$ relates the fixed, right-handed Cartesian basis $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ to the body's corotational basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, which we assume is aligned with the accelerometers' and rate gyroscopes' axes.

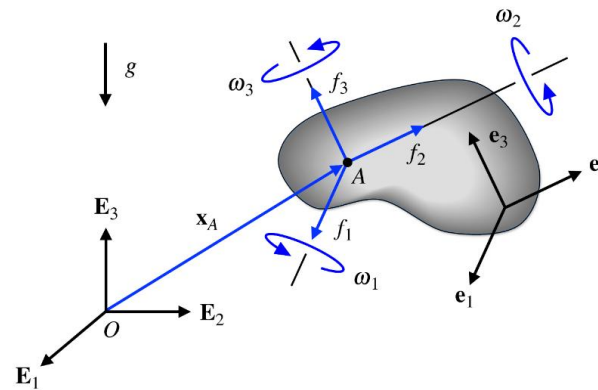


Figure 1: Schematic of a rigid body with landmark A at which an IMU is attached. The axes associated with the IMU's **accelerometers** and **rate gyroscopes** are aligned with the corotational basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$

The IMU's rate gyroscopes measure the components of the rigid body's absolute angular velocity $\boldsymbol{\omega}$ in the corotational basis. That is, the rate gyroscopes provide the signals

$$(1) \quad \omega_i(t) = \boldsymbol{\omega} \cdot \mathbf{e}_i \quad (i = 1, 2, 3).$$

The IMU's accelerometers sense the corotational components of the so-called specific force $\mathbf{f}$ exerted at the landmark A . Specific force is simply the difference between the absolute acceleration of A , $\ddot{\mathbf{x}}_A$, and gravitational acceleration, and thus the output of the accelerometers, $f_i(t)$, can be identified with the measurements

$$(2) \quad f_i(t) = \mathbf{f} \cdot \mathbf{e}_i = (\ddot{\mathbf{x}}_A - \mathbf{g}) \cdot \mathbf{e}_i.$$

Strapdown inertial navigation

IMU Strapdown Algorithm (3/4) – The navigation equations

Suppose a 3-2-1 set of Euler angles is used to parameterize $\mathbf{Q}$. In vehicle dynamics, the Euler angles ψ , θ , and ϕ are identified as the yaw, pitch, and roll angles, respectively. For this set of Euler angles, provided rate gyroscope data $\omega_i(t)$, the orientation of the rigid body is determined by integrating the following set of differential equations:

$$(3) \quad \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix} = \begin{bmatrix} 0 & \sin(\phi) \sec(\theta) & \cos(\phi) \sec(\theta) \\ 0 & \cos(\phi) & -\sin(\phi) \\ 1 & \sin(\phi) \tan(\theta) & \cos(\phi) \tan(\theta) \end{bmatrix} \begin{bmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix},$$

for which we must specify the body's initial orientation: $\psi(t_0)$, $\theta(t_0)$, and $\phi(t_0)$. You may have noticed that these equations relating the rates of change of the Euler angles and ω_i can be obtained using the dual Euler basis vectors: $\dot{\psi} = \boldsymbol{\omega} \cdot \mathbf{g}^1$, $\dot{\theta} = \boldsymbol{\omega} \cdot \mathbf{g}^2$, and $\dot{\phi} = \boldsymbol{\omega} \cdot \mathbf{g}^3$. Further, these equations are singular when $\theta = \pm \frac{\pi}{2}$ rad.

To solve for how the position vector $\mathbf{x}_A$ of the landmark A varies over time, we first transform the corotational components of specific force, $f_i(t)$, measured by the accelerometers into the fixed Cartesian frame and then correct for gravitational acceleration to obtain the fixed Cartesian components of the absolute acceleration of A , $\ddot{\mathbf{x}}_{A_i}(t) = \ddot{\mathbf{x}}_A \cdot \mathbf{E}_i$. If we consider motion near the earth's surface, then the gravitational acceleration vector $\mathbf{g} \approx -g\mathbf{E}_3$. Consequently,

$$(4) \quad \begin{bmatrix} \ddot{x}_{A_1} \\ \ddot{x}_{A_2} \\ \ddot{x}_{A_3} \end{bmatrix} = \begin{bmatrix} \cos(\psi) & -\sin(\psi) & 0 \\ \sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\phi) & -\sin(\phi) \\ 0 & \sin(\phi) & \cos(\phi) \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}.$$

Strapdown inertial navigation



IMU Strapdown Algorithm (3/4) – The navigation equations

We then twice integrate the system of differential equations (4), with initial position $x_{A_i}(t_0)$ and velocity $\dot{x}_{A_i}(t_0)$, to obtain the trajectory of A . Expressions (3) and (4) are often referred to as the navigation equations. The process we described here for computing the displacement of point A and the body's orientation over time using the IMU accelerometer and gyroscope data is summarized diagrammatically in Figure 2.

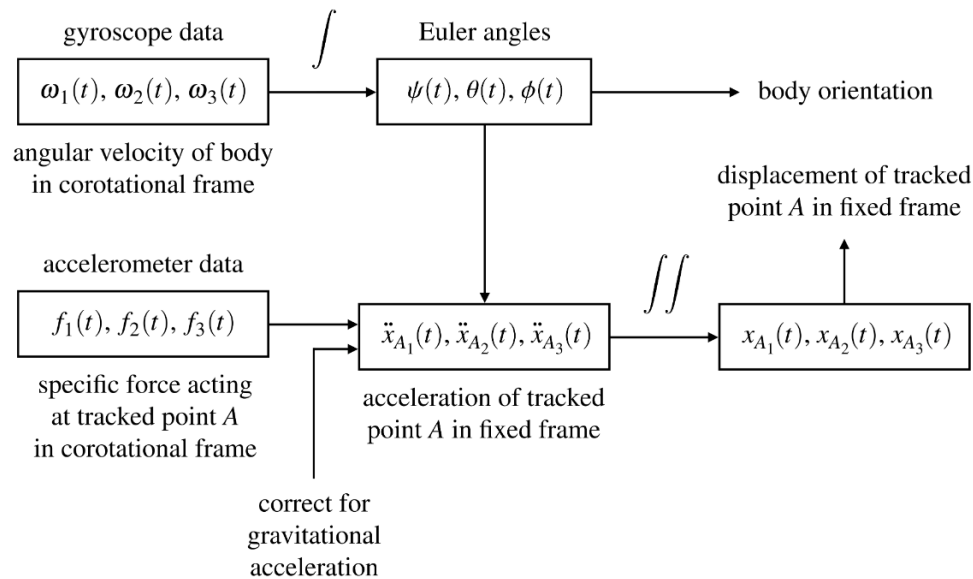


Figure 2. Diagrammatic representation of the process for calculating the displacement of a tracked point on a rigid body and the body's orientation over time from IMU accelerometer and rate gyroscope data.

Strapdown inertial navigation

IMU Strapdown Algorithm (3/4) – Integration drift and navigation error

- To illustrate the effect of integration drift, consider an IMU attached to the center A of a vehicle that travels at a constant speed in a circle of radius R_0 in the horizontal plane, as shown in Figure 3, where ψ denotes the vehicle's yaw angle

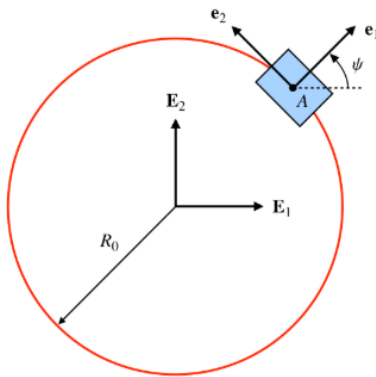


Figure 3: Top-view schematic of a vehicle illustrating the alignment of the corotational basis vectors e_1 and e_2 as the vehicle travels in a circle.

If the initial conditions for the vehicle are such that $\mathbf{x}_A(0) = R_0 \mathbf{E}_1$, $\dot{\mathbf{x}}_A(0) = \omega_0 R_0 \mathbf{E}_2$, and $\psi(0) = 0$, then the exact solutions for the position of A and the vehicle's orientation are given by

$$\begin{aligned} \mathbf{x}_A(t) &= R_0 (\cos(\omega_0 t) \mathbf{E}_1 + \sin(\omega_0 t) \mathbf{E}_2), \\ \psi(t) &= \omega_0 t. \end{aligned} \quad (5)$$

Because the vehicle's motion is confined to the horizontal plane, the navigation equations (3) and (4) reduce to

$$\begin{aligned} \dot{\psi} &= \omega_3, \\ \begin{bmatrix} \ddot{x}_{A1} \\ \ddot{x}_{A2} \end{bmatrix} &= \begin{bmatrix} \cos(\psi) & -\sin(\psi) \\ \sin(\psi) & \cos(\psi) \end{bmatrix} \begin{bmatrix} f_1 \\ f_2 \end{bmatrix}. \end{aligned} \quad (6)$$

Thus, (5) and (6) imply that the IMU must yield the following signals:

$$\begin{aligned} \omega_3(t) &= \omega_0, \\ f_1(t) &= -\omega_0^2 R_0, \\ f_2(t) &= 0. \end{aligned} \quad (7)$$

Strapdown inertial navigation



IMU Strapdown Algorithm (3/4) – Integration drift and navigation error

- Figure 4 depicts the IMU signals (7) for $R_0 = 5m$ and $\omega_0 = \pi$ rad/s.

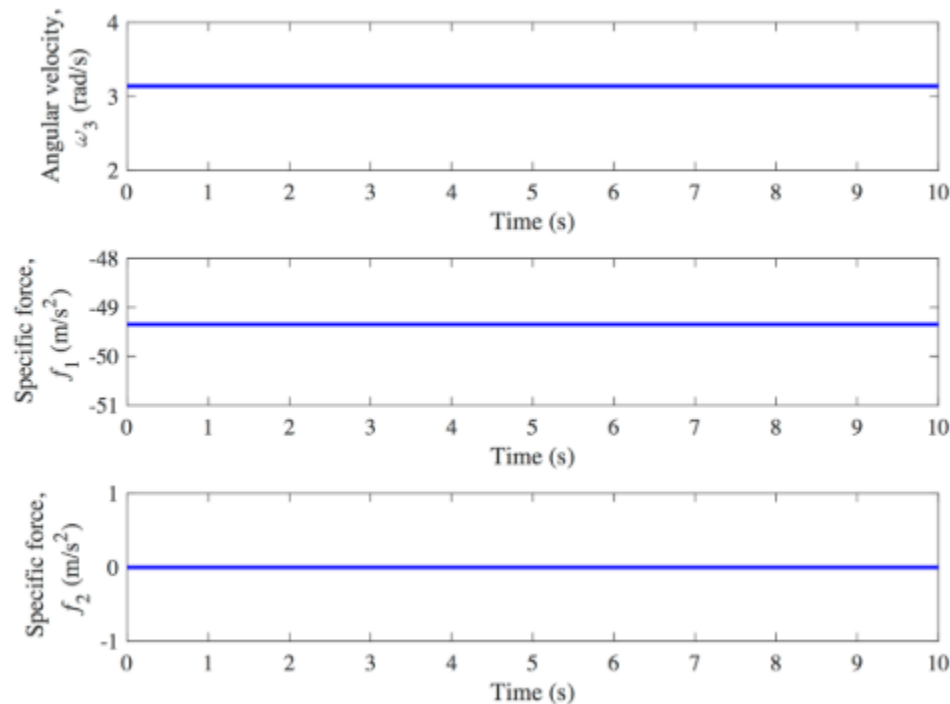


Figure 4: Noise-free angular velocity and specific force signals from an IMU attached to the center of a vehicle traveling in a circle at a constant speed.

Strapdown inertial navigation



IMU Strapdown Algorithm (3/4) – Integration drift and navigation error

- Now, suppose the IMU's sampling rate is 100 Hz. If we solve the navigation equations (6) and integrate to compute the trajectory of the vehicle's center A, we find, as shown in Figure 5, that the estimated trajectory (drawn in green) noticeably deviates from the actual circular path (drawn in red) as time progresses. This integration drift is **unavoidable**, but its effect can be slowed if the IMU were to collect data at a **higher sampling rate**

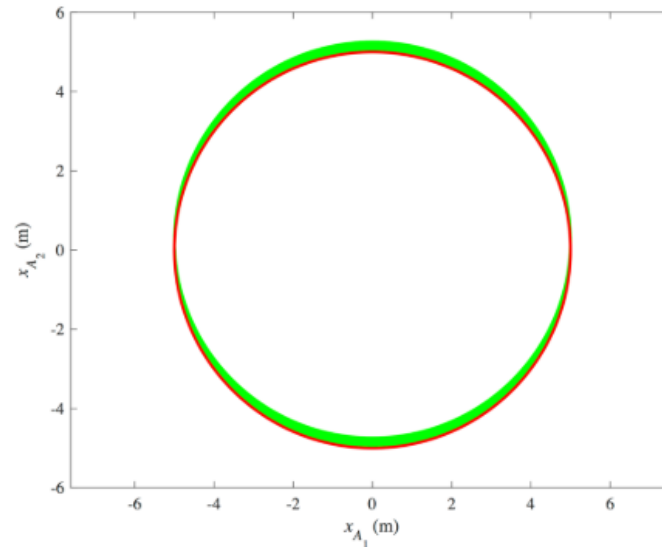


Figure 5: Illustration of integration drift for a vehicle traveling in a circle when the IMU data is free of noise. The computed trajectory of the vehicle's center is shown in green, while the actual circular path is drawn in red.

Strapdown inertial navigation



IMU Strapdown Algorithm (3/4) – Integration drift & IMU Noise

- Of course, the IMU data will be subject to noise, and therefore the angular velocity and specific force signals (7) are depicted more realistically in Figure 6.

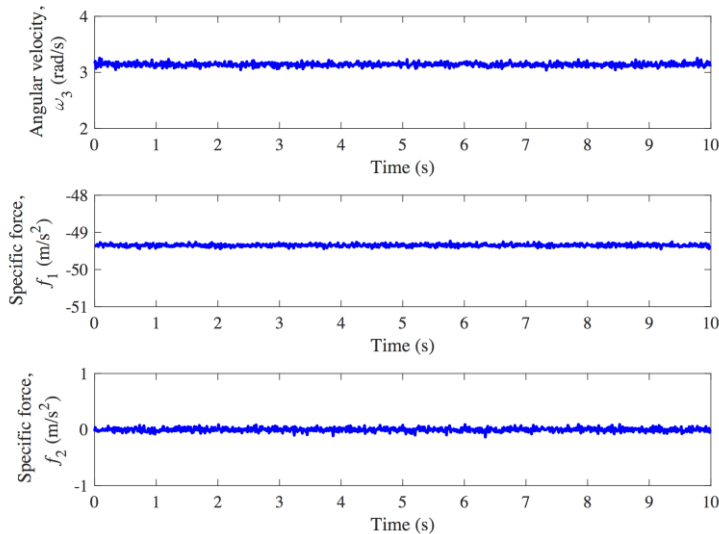


Figure 6: Noisy angular velocity and specific force signals from an IMU attached to the center of a vehicle traveling in a circle at a constant speed..

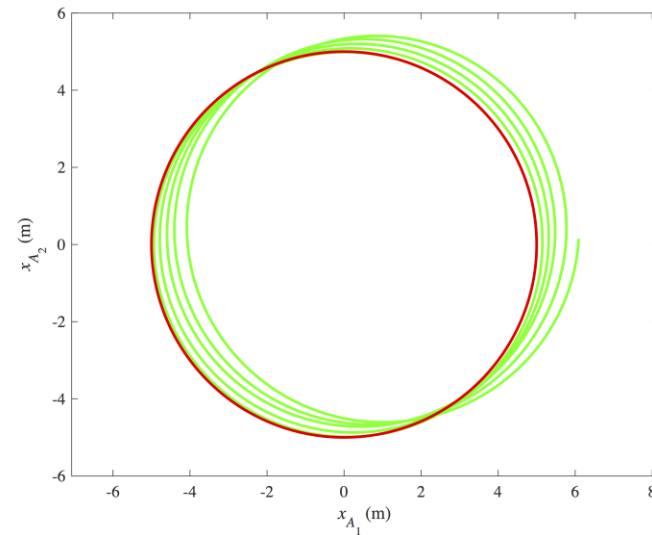
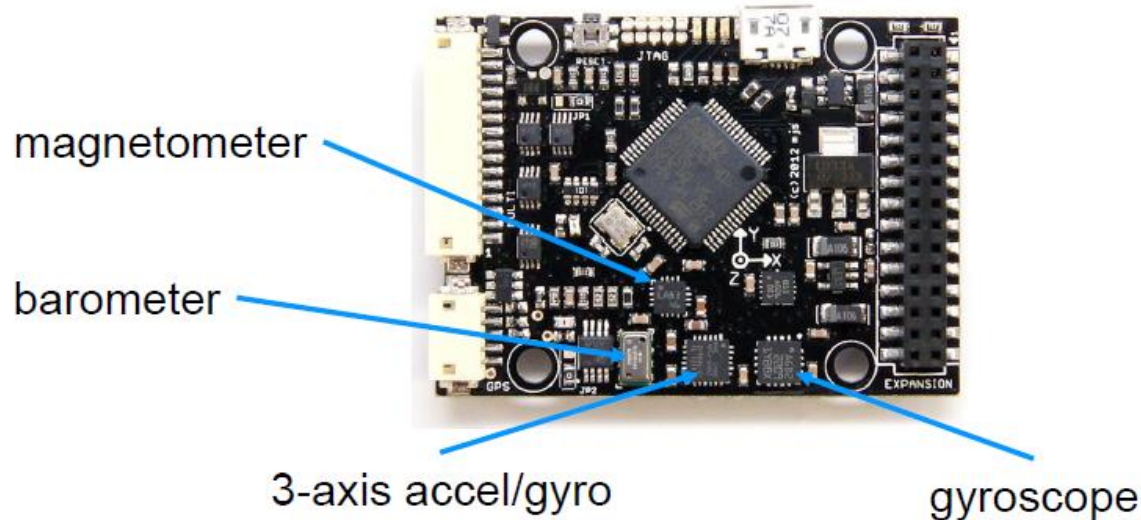


Figure 7: Illustration of integration drift for a vehicle traveling in a circle when the **IMU data is noisy**. The computed trajectory of the vehicle's center is shown in green, while the actual circular path is drawn in red.

Example using multiple Sensors- Pixhawk FMU and sensors

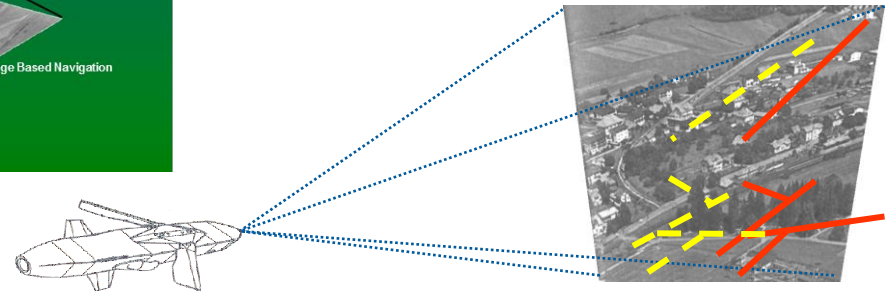
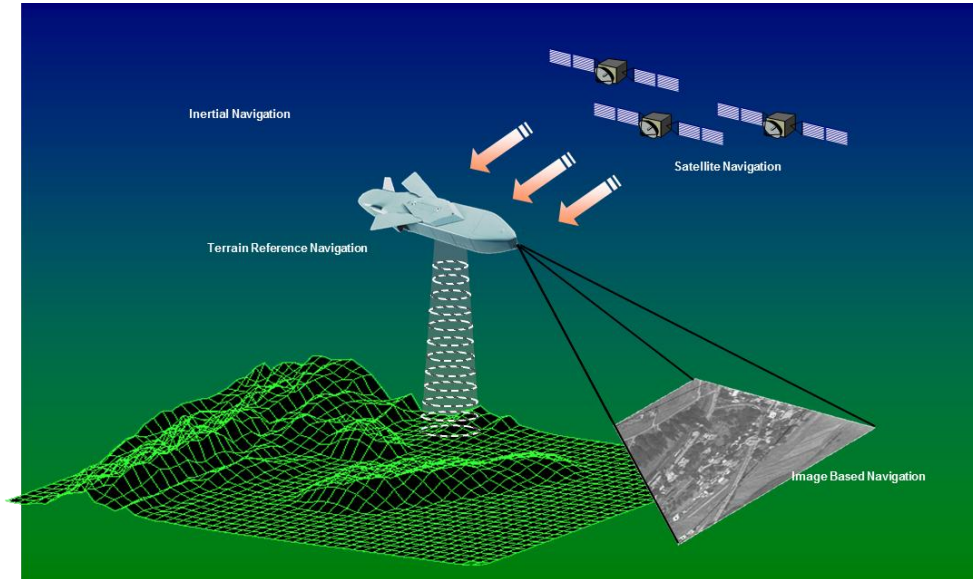
Inertial Navigation

- ARM7 Cortex-M4F microcontroller (168MHz,DSP,floating-point hardware acceleration)
- ST Micro L3GD20H 16 bit gyroscope
- ST Micro LSM303D 14 bit accelerometer / magnetometer
- InvensenseMPU 6000 3-axis accelerometer/gyroscope
- MEAS MS5611 barometer



Example using multiple Sensors- Taurus KEPD 350

Image based Navigation (Overview)



- **“TriTec” Navigation system based on Inertial Measurement Unit (IMU) and the following 3 Aiding sensors**
 - GPS (Global Positioning System)
 - TRN (Terrain Reference Navigation)
 - IBN (Image based Navigation) => New Idea AI-based